WIA1006/WID3006 MACHINE LEARNING

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TrueLove AI

Group Project Report

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Dashboard: https://wia1006-group15-dashboard.streamlit.app/

1. Problem and Objective

1.1 Background

In today's age of modernisation, online dating sites have changed the way people form relationships. People create extensive behavioural traces through the way they write their profiles and the frequency of their messaging, which can reveal their genuine intentions on these platforms. Unlike traditional relationships, modern ones often develop through digital interactions and communication via applications. This shift has given rise to phenomena like ghosting and situationships.

Although there is a wealth of user behaviour data accessible on dating platforms, not much attention has been given to classifying users by their true relationship intentions, especially in diverse communities like LGBTQ+ groups. This project aims to address that gap by using machine learning to determine whether a dating app user is looking for a serious relationship based on their behaviour and demographic signs.

1.2 Problem Statement

Among dating app users within each LGBTQ+ subgroup, what proportion exhibits behavioural patterns that indicate a serious relationship intent?

1.3 Objectives

- To engineer a meaningful target label “is_serious” from raw behavioural features using domain-informed thresholds.
- To train and compare multiple machine learning models for predicting serious relationship intent.
- To evaluate model performance across different LGBTQ+ sexual orientation groups to assess prediction fairness.
- To benchmark the manual model performance against AutoML (FLAML).

1.4 Project Significance

This project is important for several reasons. First, it shows that user behaviour on dating apps sends clear signals about relationship intent. Second, it emphasises the significance of group-aware evaluation, where a model that works well overall may still show differences among demographic groups. Lastly, the deployed dashboard offers a useful tool that extends beyond academic analysis, making machine learning predictions an interactive experience.

2. Methodology and Model Explanation

2.1 Dataset

The project references Kaggle's Dating App Behavior Dataset, which consists of 50,000 user records with 19 variables, including demographic information, app usage habits, swipe preferences, and match results. The dataset was created to be multifaceted and balanced, including category, numerical, and labelled variables.

Feature Type	Feature Columns	Description
Demographic	gender, sexual_orientation, location_type, education_level, income_bracket	User background and identity features
Behavioural	bio_length, swipe_right_ratio, message_sent_count, emoji_usage_rate, app_usage_time_min	How the user interacts with the app
Engagement	likes_received, mutual_matches, profile_pics_count	Social engagement and attractiveness signals
Outcome	match_outcome	Result of the user interaction (Mutual Match, Ghosted, etc.)

2.2 Target Label Engineering “is_serious”

The dataset does not include a clear seriousness label. The goal variable “is_serious” was created using a domain-specific grading mechanism known as the Goldilocks Rule. Each user is assigned a score depending on how many of the following criteria they satisfy. Users who score 4 or above are classified as serious daters (is_serious = 1).

No.	Criteria	Rationale
1	Bio length within ±100 of the orientation group average	Invested in self-presentation
2	Swipe right ratio below group average	Selective, not swiping indiscriminately
3	Message count above group average	Genuinely engaged in conversations
4	Emoji usage within ±0.15 of the group average	Natural communication style
5	Message density > 0.5 messages per minute	Actively spending quality time talking
6	Match ROI > 5 (mutual matches/swipe ratio)	Selective swiping that leads to real connections
7	Outcome is Relationship Formed, Blocked or No Action	Red flag penalty, disengagement or no follow-through

Thresholds are computed per sexual orientation group using group-specific averages and not globally. Each user is compared against their own community's behaviour to avoid majority group prejudice and maintain fairness.

2.3 Data Preprocessing

- **Missing value handling:** Numeric columns were filled with the median, while categorical columns were filled with the mode.
- **Feature engineering:** interest_tags was converted into a numeric interest_count feature by counting comma-separated values.
- **Encoding:** sexual_orientation was one-hot encoded using scikit-learn's OneHotEncoder with default settings. Numeric features were standardised using StandardScaler via a ColumnTransformer pipeline.
- **9 core features selected**: 'bio_length', 'likes_received', 'app_usage_time_min', 'message_sent_count', 'emoji_usage_rate', 'swipe_right_ratio', 'mutual_matches', 'profile_pics_count','sexual_orientation'.
- **Train & test split:** 80% training, 20% testing with random_state=42.

2.4 Models

Six machine learning models were developed and assessed. To guarantee a fair comparison, all models used the same preprocessed training data, train/test split, and evaluation metrics.

Model	Key Configuration	Characteristics
Random Forest	n_estimators=100, max_depth=15, random_state=42	Ensemble of decision trees: robust to noise and imbalance
Neural Network (MLP)	hidden_layer_sizes=(64,32), max_iter=1000, random_state=42	Deep learning: captures non-linear patterns
Decision Tree	random_state=42	Interpretable single tree structure with clear decision rules
K-Nearest Neighbours	n_neighbors=5	Distance-based: sensitive to feature scaling
Logistic Regression	max_iter=1000, random_state=42	Linear statistical baseline: fast and interpretable
SVM	LinearSVC(dual=False, max_iter=2000, random_state=42) kernel='rbf', class_weight='balanced'	Linear kernel SVM: effective for high-dimensional data

2.5 Evaluation Metrics

Accuracy alone is inadequate as an essential metric given the dataset's class imbalance (66.8% casual vs. 33.2% serious users), where a model that predicts everyone as casual will obtain 62% accuracy without learning anything significant. Therefore, the following metrics were used:

- **F1-Score (primary):** Harmonic mean of precision and recall. Penalises both missed serious users and false alarms. Used as the primary ranking metric.
- **Accuracy:** Overall proportion of correct predictions. Reported alongside F1 for completeness.
- **Precision:** Of all users predicted as serious, how many were actually serious.
- **Recall:** Of all actual serious users, how many were correctly identified.
- **AutoML benchmark:** FLAML AutoML was run for 60 seconds to optimize for F1-score, and its best model was compared against our manual models.

3. Results and Visualisation

3.1 Target Label Distribution

After applying the Goldilocks Rule across all 50,000 users, the resulting "is_serious" distribution was as follows:

Class	Count (%)
Casual (0)	33,376 (66.8%)
Serious (1)	16,624 (33.2%)

The class imbalance is moderate, approximately 66.8 casual and 33.2 is serious. This project trained all models directly on the natural class distribution. F1-score was prioritised as the evaluation metric to account for this imbalance.

3.2 Model Performance Results

All six models were trained on 40,000 samples and evaluated on the real-world test set of 10,000 samples. Results are ranked by F1-Score:

Rank	Model	Accuracy	F1-Score	Precision	Recall
1	Random Forest ★	90.55%	0.8631	0.83	0.90
2	Neural Network	89.80%	0.8511	0.82	0.88
3	Decision Tree	87.32%	0.8052	0.82	0.79
4	KNN	81.15%	0.7116	0.72	0.70
5	Logistic Regression	77.22%	0.6251	0.68	0.58
6	SVM	77.21%	0.6249	0.68	0.58

Random Forest achieved the highest F1-Score of 0.8631 and accuracy of 90.55%, making it the clear best model. Neural Network was a close second at 0.8511 for F1-score. Logistic Regression and SVM performed similarly with an F1-score of approximately 0.625, reflecting their limitations in handling non-linear decision boundaries. KNN struggled, achieving a moderate F1 of 0.7116 due to its sensitivity to the class imbalance.

3.3 AutoML Benchmark (FLAML)

FLAML AutoML was run for 60 seconds to optimize for F1-Score. It automatically searched across multiple model types and hyperparameter combinations. The best model identified by FLAML was compared against our top manually trained model:

Method	Accuracy	F1-Score
Manual Best (Random Forest)	90.55%	0.8631
FLAML AutoML	91.04%	0.8727

FLAML confirmed that Random Forest is close to the optimal automated choice for this dataset, validating our model selection approach.

3.4 LGBTQ+ Group Analysis

Serious dater rates varied across sexual orientation groups, indicating variances in community behaviour standards.

Orientation	Casual Rate	Serious Rate
Lesbian	66.4%	33.6%
Demisexual	67.0%	33.0%
Gay	67.3%	32.7%
Queer	67.2%	32.8%
Straight	66.8%	33.2%
Bisexual	66.4%	33.6%
Pansexual	66.5%	33.5%
Asexual	66.3%	33.7%

Asexual users showed the highest serious dater rate at 33.7%, while gay users showed the lowest rate at 32.7%. These variations are minor, within only 1% point, suggesting that sexual orientation is not a reliable and strong predictor of seriousness. The model focuses more on real behavioural cues.

4. Insights and Interpretation

4.1 The Outperformance of the Random Forest Model

- **Ensemble robustness:** Random Forest decreases variance and prevents overfitting by aggregating 100 decision trees, each trained on random feature subsets.
- **Non-linearity:** The relationship between seriousness and behavioural features is not linear. Random Forest captures complex interactions between features that Logistic Regression and SVM with linear kernels cannot.
- **Feature importance:** Random Forest identifies relationships among features, such as the combination of a high message count and selective swiping, which are both required to signify seriousness.
- **Depth constraint:** Setting max_depth=15 prevents overfitting while maintaining sufficient model complexity.

4.2 The Reason Other Models Underperformed

- **KNN:** After one-hot encoding, the feature space was expanded. Distance calculations become less meaningful in high-dimensional spaces because it is sensitive to the curse of dimensionality.
- **Logistic Regression&SVM:** Both assume linear decision boundaries and scored similarly at approximately 0.625. The Goldilocks Rule develops a nonlinear metric that linear models fail to sufficiently capture.
- **Neural Network:** The F1-score of 0.8511, demonstrating strong non-linear learning. The gap from Random Forest is small, suggesting the dataset does not require very deep representation learning.
- **Decision Tree:** The tree somewhat overfits the training data in the absence of a depth constraint. Despite good accuracy at 87.32%, the F1-score of 0.8052 shows that it missed more serious users than Random Forest.

4.3 Goldilocks Rule Design Insights

The orientation-aware thresholds approach was a key design decision that produced several advantages:

- **Prevents majority group bias**: The global thresholds would be dominated by the largest or most common orientation group.
- **Community-relative fairness:** Each user is evaluated against their own orientation group's norms, ensuring no group is disadvantaged by thresholds shaped by another group's behaviour.
- **Aligns with real-world intuition:** what constitutes normal or selective behaviour differs across communities.

The grading method requires a minimum score of 4 out of 6 to be considered serious, allowing for considerable flexibility. Users do not need to satisfy all the requirements to be classified as serious. This simulates real-world complexity, where a person may be very engaged in messaging yet have a shorter bio or use fewer emojis, while being truly serious.

4.4 LGBTQ+ Fairness Observations

The variance in serious dater rates among sexual orientation groups is modest. However, particular observations are notable:

- Asexual users showed the highest serious dater rate at 33.7%, followed by Bisexual and Lesbian, both 33.6%. This is likely due to the above-average message counts and longer bios in these communities.
- Gay users showed the lowest rate at 32.7%, which is a negligible difference. This strongly suggests that seriousness is driven by individual behaviour rather than sexual orientation.
- The small range across orientation groups suggests the model generalises well across LGBTQ+ communities without significant group-level bias.
- Using orientation-aware thresholds means the model does not systematically disadvantage any group based on their community's average behaviour patterns.

4.5 Key Behavioural Predictors

Based on Random Forest feature importance and the Goldilocks scoring analysis, the most influential features for predicting seriousness were:

- **Message count:** The strongest single predictor. Users who contribute above-average messages for their group demonstrate real interest.
- **Swipe right ratio:** Selective swipers are more likely to be serious. They are not using the app casually.
- **Bio length:** Spending time in a well-crafted bio demonstrates true self-presentation purpose.
- **Emoji usage rate:** Natural emoji usage correlates with an authentic communication style.
- **Message density:** High messages-per-minute suggests quality engagement rather than passive app usage.

5. Conclusion

5.1 Summary

This project successfully demonstrated that behavioural signals on dating apps can predict serious relationship intent. By engineering the “is_serious” target label using an orientation-aware Goldilocks Rule across 50,000 user records, and training six machine learning models on 40,000 samples evaluated against 10,000 test samples, the following was achieved:

- Random Forest achieved the best performance with 90.55% accuracy and an F1-score of 0.8631, outperforming all other models, including AutoML.
- The Goldilocks Rule produced a meaningful, fair target label by using community-specific thresholds rather than a one-size-fits-all approach.
- Serious dater rates varied within just one percentage point across all LGBTQ+ groups (32.7%–33.7%), confirming that seriousness is primarily driven by individual behaviour rather than sexual orientation.
- An interactive Streamlit dashboard was deployed, providing live predictions and statistical visualisations.

5.2 Limitations

- **Engineered target:** “is_serious” is a proxy label based on behavioural thresholds, not ground truth. Different threshold choices would produce different labels.
- **No class balancing:** Models were trained using a natural class distribution (66.8% casual, 33.2% serious). This may result in models underperforming on a serious minority class in certain cases.
- **Static model:** The model does not update over time as user behaviour evolves.

5.3 Closing Remarks

The project showed that digital behaviour on dating sites is a reliable indicator of relationship intent. Our team created a comprehensive machine learning pipeline that takes raw data from raw data to a real-world application by integrating domain-driven feature engineering, robust multi-model comparison, and an interactive dashboard. The modest variance in serious dater rates among LGBTQ+ groups is particularly remarkable, implying that what defines a serious dater beyond sexual orientation is founded in universal behavioural patterns.